Quadratic-width learning with symmetry-invariant neuron merging

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Abstract

We strengthen a quadratic-width Bayesian learning theorem by removing two artifacts of its neuron-merging restriction: dependence on arbitrary hidden neuron signs and the requirement of nearly equal population sizes. The new condition allows signed populations of arbitrary sizes, with different population counts in different layers. A mass-normalized representation makes every compressed Gaussian coordinate have the same variance, independently of the smallest population. A scale-normalized readout budget controls the number of populations. The resulting sample bound remains quadratic in Boolean teacher width, up to logarithms, with polynomial depth dependence, at every learner width and every specified Gaussian scale at or above mean field. We also give a local operator-norm certificate: approximate agreement of rows on population-constant states, together with internal-noise robustness, implies prediction-preserving merging. These are structural prior restrictions; the paper does not establish that ordinary trained networks satisfy them. The results are not Lean verified.

1 Which aspects of neuron merging are natural?

Clustering redundant neurons is a plausible way to reduce a large network to a smaller representation. A theorem based on this idea should respect exact symmetries of the network, and should not require every cluster to have almost the same size. Both requirements matter mathematically.

The previous population-averaging condition can distinguish two identical functions solely because one hidden neuron has been sign-flipped. Take $d = 1$, $N = 2$, one hidden layer, first-layer rows $(2, 0), (2, 0)$, and readout $(3/8, 3/8)$. The output is $(3/4)\tanh(2x)$ and $E_v = N\|v\|_2^2 = 9/16$. The previous budget $q \le 1 + E_v$ permits only one population, whose averaging preserves this network exactly. Flip the second row and its readout sign. The function and Gaussian density are unchanged, but one-population averaging now gives the zero function, changing clipped scores by $1/2$. Both representations satisfy the internal-noise condition below at $c = 0.1$: the calibrated ball changes their raw outputs by at most 0.1, and the clean output magnitude is greater than 0.7.

Allowing signed populations repairs this defect. Allowing arbitrary group sizes is more subtle: ordinary compressed weights can become large when a population is small. We remove this difficulty by rescaling compressed hidden states by the square root of their population mass. The resulting activation family is uniformly Lipschitz in its scale parameter, so even very small populations admit a finite code independent of learner width.

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The final section gives a local sufficient condition for merging. It uses operator norms of explicit weight defects instead of assuming directly that predictions are preserved. The empirical prevalence of either condition remains an open question.

2 Model and a symmetry-respecting support condition

Let $H = L - 1 \geq 1$, $d \geq 1$, and $\mathcal{X} = \{-1, 1\}^d$. The learner has H hidden tanh layers of width N, first-layer augmented input $\bar{x} = (x, 1)$, and linear readout:

$$h^1 = \tanh(W^1 \bar{x}), \qquad h^s = \tanh(W^s h^{s-1}), \qquad z_w(x) = v^\top h^H.$$

Under $\mu_{N,t}$, independent Gaussian entries have variances t^2/d, t^2/N, and t^2/N^2 in the first layer, subsequent hidden layers, and readout, respectively; $0 < t \leq 1$. Taking $t = N^{(1-\gamma)/2}$ includes every $\gamma \geq 1$. Set

$$E_v = N \|v\|_2^2, \qquad M(w) = \max_{x \in \mathcal{X}} |z_w(x)|, \qquad T(u) = \mathrm{clip}(u, -1/2, 1/2).$$

At hidden layer s, choose a partition into q_s nonempty groups of arbitrary sizes $k_{s,a}$ and a sign $\sigma_i \in \{-1, 1\}$ for each neuron. Let F_s have column a equal to the signed normalized indicator of that population: its entries there are $\sigma_i / \sqrt{k_{s,a}}$. Then $F_s^\top F_s = I$, and $R_s = F_s F_s^\top$ averages neurons after aligning their signs. Define the merged network by

$$\bar{W}^1 = R_1 W^1, \qquad \bar{W}^s = R_s W^s R_{s-1}, \qquad \bar{v} = R_H v. \tag{1}$$

Write $q = \max_s q_s$. The new event $\mathcal{A}_N$ requires the existence of such signed populations satisfying

$$(q-1) M(w)^2 \leq 4 E_v(w), \qquad \max_x |T(z_w(x)) - T(z_{\bar{w}}(x))| \leq 1/16. \tag{2}$$

The division-free budget includes $M = 0$ without a special convention. It is invariant under readout rescaling. The clipped-score comparison uses fixed margins and retains their output-scale dependence. The complete condition is invariant under hidden-neuron sign flips and permutations. Indeed, a sign change S_s replaces F_s by $S_s F_s$ and preserves all functions and energies in the condition. All witnesses are finite choices at fixed N, so the event is Borel.

We can additionally impose internal-noise robustness. Add disturbances after every activation and at the output, with

$$\|\eta\|_*^2 = \sum_s \|\eta^s\|_2^2 / N + |\eta^{\mathrm{out}}|^2, \qquad r_c(w) = c / \sqrt{1 + H E_v(w)}, \quad c > 0.$$

Let $\mathcal{R}_{N,c}$ require

$$|T(z_w(x; \eta)) - T(z_w(x))| \leq 1/16 \quad (x \in \mathcal{X},\ \|\eta\|_* \leq r_c(w)). \tag{3}$$

The prior is the raw Gaussian conditioned on $\mathcal{A}_N$, optionally also on $\mathcal{R}_{N,c}$. Training further conditions on $f(X_i) z_w(X_i) > 1/2$. All statements concern exact Bayesian conditioning.

3 An exact normalized representation

Put $a_{s,b} = \sqrt{k_{s,b}/N}$ and define

$$B^1 = F_1^\top W^1/\sqrt{N}, \qquad B^s = F_s^\top W^s F_{s-1}, \qquad b = \sqrt{N} F_H^\top v.$$

The merged network has exactly the following compressed representation:

$$g^1 = \phi_{a_1}(B^1 \bar{x}), \qquad g^s = \phi_{a_s}(B^s g^{s-1}), \qquad z_{\bar{w}} = b^\top g^H, \tag{4}$$

where the activation is applied coordinatewise and

$$\phi_a(u) = a \tanh(u/a) \quad (a > 0), \qquad \phi_0(u) = 0.$$

To verify the identity, the full hidden state is $\sqrt{N} F_s g^s$; oddness of tanh gives (4) at every layer.

Lemma 1 (Lipschitz scale coordinates)**.** *For $a, b \geq 0$ and real u, v,*

$$|\phi_a(u) - \phi_a(v)| \leq |u - v|, \qquad |\phi_a(u) - \phi_b(u)| \leq |a - b|.$$

Proof. The input derivative is $\operatorname{sech}^2(u/a)$. The scale derivative is $\tanh y - y \operatorname{sech}^2 y$, where $y = u/a$. It is odd and, for $y \geq 0$, increases from 0 to 1: its derivative is $2y \operatorname{sech}^2 y \tanh y \geq 0$. Continuity at $a = 0$ follows from $|\phi_a(u)| \leq a$. □

For fixed populations, every entry of $\sqrt{d} B^1$, B^s for $s \geq 2$, and b is an independent Gaussian of variance exactly t^2/N. This follows from the orthonormal columns of F_s, and is independent of the group sizes. Write

$$\kappa = d \left\| B^1 \right\|_F^2 + \sum_{s=2}^{H} \|B^s\|_F^2 + \|b\|_2^2, \quad P = q_1(d+1) + \sum_{s=2}^{H} q_s q_{s-1} + q_H.$$

Then

$$\mathbb{E} e^{N\kappa/(4t^2)} = 2^{P/2}, \qquad P/N \leq d + 2 + (H-1)Q \quad (q_s \leq \min(N, Q)). \tag{5}$$

The auxiliary representation uses perspective activations; the learner itself continues to use ordinary tanh.

4 A quadratic sample bound

The Boolean teacher has width C and depth H, with NAND, copy, and negation gates. Constant wires and basis-compilation overhead count toward these resources. Define

$$\begin{aligned}
e_* &= (16\sqrt{2C})^{-1}, & \tau_0 &= e_*/4, \\
b_L &= \sqrt{16 \log 9 + 8 \log(4L)}, & A &= \max\{32 b_L, 2 \log(8/e_*)\}, \\
a &= \max\{4, 256c\}, & \rho &= \min\{1/4, A/[4(d+1)], A e_*/(64\sqrt{C})\}, \\
q_0 &= d + 2 + (H-1)C, & B_0 &= 9dA^2 + 18(H-1)CA^2 + 2a^2 C, \\
D_* &= B_0 + q_0 \log \frac{e\sqrt{2\pi}}{2\rho} + \log 2.
\end{aligned} \tag{6}$$

Appendix A proves that the target has a cylinder of raw probability at least e^{-ND_*/t^2} in the stated support, for every $N \geq C$. In particular $D_* = O((d + LC)\log^2(2dLC) + c^2C)$.

For $0 < \varepsilon, \delta < 1$, set

$$\begin{aligned}
&B = \max\{8, 8[D_* + \log(4/\varepsilon)]\}, \qquad Q = \lceil 1 + 16B \rceil, \\
&J = 4\left[D_* + \log(4/\varepsilon) + H\log Q + H\log(2Q) + \tfrac{\log 2}{2}\{d + 2 + (H-1)Q\}\right], \\
&R = 1 + \sqrt{J}, \qquad A_g = 2 + d + (Q+1)(R+1), \\
&\zeta = [32(H+1)A_g^{H+1}]^{-1}, \qquad D_Q = Q(d+2) + (H-1)Q^2 + HQ.
\end{aligned} \tag{7}$$

Theorem 2. *If*

$$n \geq \frac{2}{\varepsilon}[D_Q \log(3 + 4R/\zeta) + \log(1/\delta)],$$

then, with probability at least $1 - \delta$ over iid inputs from any fixed distribution D, the posterior satisfies

$$\mathbb{E}_{w \sim Q_{S,N,t,c}} \mathbb{P}_{X \sim D}\{f(X) z_w(X) \leq 1/4\} \leq \varepsilon$$

simultaneously for every $N \geq C$ and $0 < t \leq 1$. All normalizers are positive. The theorem applies with or without the additional condition $\mathcal{R}_{N,c}$.

Proof. The target cylinder lies in every noiseless training event, so the posterior normalizer relative to the raw Gaussian is at least e^{-ND_*/t^2}. Since NE_v/t^2 is chi-square with N degrees of freedom,

$$\mu_{N,t}(E_v > B) \leq e^{-NB/(8t^2)} \quad (B \geq 8).$$

After normalization this event has probability at most $\varepsilon/4$. A nonempty successful training sample forces $M > 1/2$; therefore (2) implies $q < 1 + 16E_v$. On the retained posterior, all witnessing population counts are at most Q.

There are at most $Q^H(2Q)^{HN}$ tuples of signed partitions with each $q_s \leq \min(N, Q)$. From (5), the raw probability that any such tuple has $\kappa > J$ is at most

$$\exp\left(H\log Q + HN\log(2Q) + \tfrac{N}{2}[d + 2 + (H-1)Q]\log 2 - \frac{NJ}{4t^2}\right).$$

The choice of J, using $N/t^2 \geq 1$ and $t \leq 1$, bounds this by $e^{-ND_*/t^2} e^{-(N/t^2)\log(4/\varepsilon)}$. Its posterior probability is at most $\varepsilon/4$. This is a union over all partitions before selecting a witness from the weights.

Every retained compressed coefficient is bounded by $\sqrt{J}$, and every scale is in $[0, 1]$. Pad to width Q with zero scales. Round coefficients and scales to fixed grids with error at most ζ, preserving zero and the interval $[0, 1]$ for scales. Lemma 1 gives hidden coordinate errors

$$e_1 \leq (d+2)\zeta, \qquad e_s \leq QRe_{s-1} + (Q+1)\zeta.$$

The output error is at most $QRe_H + Q\zeta \leq 1/32$. These rounded functions form a class $\mathcal{G}$, independent of N, t, with

$$\log|\mathcal{G}| \leq D_Q \log(3 + 4R/\zeta).$$

The score difference from the original predictor is at most $1/16 + 1/32 = 3/32$.

Use a ramp loss on $f(x)T(h(x))$ equal to one below $11/32$, zero above $13/32$, and affine between these thresholds. Each retained predictor has a candidate in $\mathcal{G}$ with zero empirical ramp loss whose

population ramp loss bounds the original predictor's test-margin error. A finite-class zero-loss bound gives, with probability at least $1-\delta$,

$$\mathbb{P}_D\{fz_w \le 1/4\} \le [\log|\mathcal{G}| + \log(1/\delta)]/n$$

for every retained predictor. For completeness, a fixed $[0,1]$-valued loss of mean r is zero on all n examples with probability at most $(1-r)^n \le e^{-nr}$; then union bound. Add the discarded posterior mass $\varepsilon/2$. The candidate class, and hence this sample event, is independent of learner width and Gaussian scale. □

The sample rate is

$$\widetilde{O}\left(\frac{LdQ + L^2Q^2 + \log(1/\delta)}{\varepsilon}\right), \qquad Q = \widetilde{O}(d + LC + c^2C + \log(1/\varepsilon)).$$

For a single-output bounded-fan-in teacher, pruning the output's ancestors gives width $C' \le \min(C, 2^{H-1}+1)$, including at most one constant wire per stage. Thus $\log C' = O(L)$. To make the polynomial degree explicit, put $S = 2 + d + L + c + \log(4/\varepsilon)$. The cylinder parameters satisfy $A, a = O(S)$ and $\log(1/\rho) = O(S)$; hence

$$D_* = O(C'S^3), \quad Q = O(C'S^3), \quad J = O(C'S^4), \quad D_Q = O(C'^2S^7),$$

while $\log(3 + 4R/\zeta) = O(S^2)$. Consequently there is an absolute constant K_0 for which

$$n \ge \frac{K_0}{\varepsilon}[C^2(2 + d + L + c + \log(4/\varepsilon))^9 + \log(1/\delta)]$$

is sufficient. The extra HQ scale parameters do not change this degree.

5 A local weight certificate for merging

Signed population projections commute with the property of being a tanh hidden state: applying tanh to a vector in their range keeps it in their range. For proposed signed projections R_s, define

$$\Delta_1 = \sqrt{(d+1)/N}\left\|(I - R_1)W^1\right\|_{\mathrm{op}}, \qquad \Delta_s = \|(I - R_s)W^s R_{s-1}\|_{\mathrm{op}} \quad (s \ge 2).$$

These measure disagreement of rows after restricting their inputs to population-constant states. They permit arbitrary weights on directions orthogonal to the preceding populations.

Proposition 3 (Local agreement implies prediction stability)**.** *If $w \in \mathcal{R}_{N,c}$ and $\sum_s \Delta_s^2 \le r_c(w)^2$, then the merged network obeys the score comparison in (2).*

Proof. Run the merged network. Its preceding hidden state lies in the range of R_{s-1} and has norm at most $\sqrt{N}$. The difference between its next hidden state and $\tanh(W^s\bar{h}^{s-1})$ has RMS norm at most Δ_s, by tanh's Lipschitz property. The first-layer bound follows from $\|\bar{x}\|_2 = \sqrt{d+1}$. Thus the complete merged trajectory is the original network under postactivation disturbances of total norm at most $(\sum_s \Delta_s^2)^{1/2}$. The readout agrees because the last hidden state lies in the range of R_H. Apply (3). □

Consequently we may replace the direct output-comparison support condition by the existence of populations satisfying the same count budget and this local certificate, together with $\mathcal{R}_{N,c}$. The target cylinder still has enough mass. In (6), shrink ρ further to

$$\rho \le \frac{c}{2\sqrt{1+4Ha^2C}} \left[\frac{(d+1)^2}{d} + (H-1)C \right]^{-1/2}. \tag{8}$$

On the cylinder, $E_v \le 4a^2C$, $\Delta_1 \le \rho(d+1)/\sqrt{d}$, and $\Delta_s \le \rho\sqrt{C}$ for $s \ge 2$. Indeed, the teacher is exactly population-constant, and the Gaussian bulk annihilates the previous population subspace. Equation (8) therefore gives the local certificate uniformly throughout the cylinder.

Recompute D_* with this smaller radius. Theorem 2 applies unchanged. The additional mass cost is logarithmic in the reciprocal radius:

$$D_* = O\left((d+LC)[\log^2(2dLC) + \log(2+c+c^{-1})] + c^2C \right).$$

Thus width dependence remains quadratic up to logarithms. A literal degree-nine envelope also holds after adding c^{-1} to the displayed parameter sum. This condition is a concrete approximate agreement of block row sums, sometimes called an approximate equitable partition or lumpability condition. It is still restrictive, and no claim about its prevalence in trained networks follows from the theorem.

6 Other calibrations and remaining questions

A purely parameter-based alternative replaces $M(w)$ by $\|v\|_1$:

$$(q-1)\|v\|_1^2 \le 4N\|v\|_2^2.$$

The same theorem and constants apply. The target cylinder has $\|v\|_1 \le a+\rho$, and any successful nonempty training sample forces $\|v\|_1 > 1/2$. This condition is stronger than the M-normalized one. It uses the standard effective support size $\|v\|_1^2 / \|v\|_2^2$: the population budget is bounded by four times the inverse effective fraction of readout neurons, plus one. It can reject diffuse dense readouts, so easier evaluation should not be confused with empirical naturality.

The original unnormalized count budget $q \le 1 + E_v$ is also compatible with signed arbitrary-size populations: replace Q by $\lceil 1+B \rceil$ and leave the remaining proof unchanged.

Why ordinary dropout needs a separate argument. Stability under neuron removal is another natural candidate, but the present target cylinder does not automatically establish it at mean-field scale. For example, let a Gaussian bulk row satisfy

$$E \sim \mathcal{N}\left(0, \frac{t^2}{N}\left[I - \frac{\mathbf{1}\mathbf{1}^\top}{N}\right]\right).$$

Its clean action on the constant state is exactly zero. If m uniform coordinates are sampled with replacement and rescaled by N/m, the resulting random state a instead has

$$\mathbb{E}(Ea)^2 = t^2(N-1)/m.$$

Indeed, its multinomial count vector gives $\mathbb{E}\|a\|_2^2 = N + N(N-1)/m$ and $\mathbf{1}^\top a = N$. Sampling without replacement gives $t^2(N/m-1)$. Thus a fixed retained width can amplify Gaussian bulk

noise as N grows. This obstructs a direct transfer of the existing cylinder proof to dropout stability; it is not a theorem ruling out every dropout prior or learning guarantee.

These extensions remove exact representation artifacts. They do not prove that an arbitrary robust network has a small population description. Small activation distances on a training sample do not by themselves imply the uniform condition, and arbitrary soft projections need not preserve tanh's nonlinear structure. The local certificate separates one sufficient weight condition from these unresolved empirical and structural questions.

Acknowledgments and verification. This note builds on the earlier collaborative robust-prior construction, including contributions from Claude. Independent AI-agent checks examined the sign counterexample, mass normalization, Gaussian partition bound, scale-sensitive grid, and local certificate. This is a research draft, not a Lean-verified theorem or a claim of empirical validation.

A A target implementation of sufficient prior mass

Lemma 4 (Stable Boolean cylinder)**.** *For every $N \geq C$ and $0 < t \leq 1$, there is a set of weights contained in $\mathcal{A}_N \cap \mathcal{R}_{N,c}$ on which $f(x)z_w(x) > 1$ for every input, with raw Gaussian probability at least $\exp(-ND_*/t^2)$.*

Proof. Partition the neurons into C balanced clone groups of sizes k_b. Then $C/2 \leq N/k_b \leq 2C$. Fill unused logical slots with constants or copies, so the ideal Boolean state $b^s(x)$ has entries in $\{-1, 1\}$, is constant on each group, and has norm $\sqrt{N}$. Write P for averaging within the clone groups.

Use gain A in every gate. NAND has signed affine expression $1 - u - v$; copies and negations use u and $-u$. Count constant wires in internal layers. Each ideal signed preactivation is at least A. Divide each coefficient from a logical wire equally among its clones, and let the readout average the output-wire clones with total coefficient a. Denote these teacher weights by w_*.

In coordinates $\sqrt{d}W^1$, $\sqrt{N}W^s$, and Nv, all raw entries have variance t^2. Restrict every first-layer coordinate, every internal row's coordinate along each normalized clone-group direction, and every readout coordinate to within ρt of its teacher value. Internal matrices then have the form

$$W^s = W^s_* + R^s + E^s, \qquad R^s = R^s P, \quad \|R^s\|_{\mathrm{op}} \leq \rho t\sqrt{C}, \quad E^s P = 0.$$

The unconstrained E^s are independent Gaussian bulk matrices. Two $1/4$-nets, each of cardinality at most 9^N, give

$$\mathbb{P}\{\|E^s\|_{\mathrm{op}} > b_L t\} \leq 2\exp(2N\log 9 - Nb_L^2/8) \leq \frac{1}{2L}.$$

Indeed, each fixed bilinear form has variance at most t^2/N, and the operator norm is at most twice its maximum over the two nets. All bulk norm bounds therefore hold with conditional probability at least $1/2$.

We check finite disturbance stability, allowing RMS disturbance τ_0 separately at each hidden layer and scalar disturbance at most τ_0 at the output. This contains the total-budget ball $\|\eta\|_* \leq \tau_0$. Put $e_s = \left\|h^s_\eta - b^s\right\|_2/\sqrt{N}$. If $e_{s-1} \leq e_*$, each group mean differs from its ideal bit by at most $e_*\sqrt{N/k_b} \leq 1/16$. At most three teacher coefficients enter a gate, so its intended signed preactivation remains at least $A/2$. For $\xi \in \{-1, 1\}$ the elementary bound

$$\xi q \geq A/2 \quad \Longrightarrow \quad |\tanh(q + u) - \xi| \leq 2e^{-A/2} + 8|u|/A$$

follows by considering $|u| < A/4$ and its complement. Since $E^s b^{s-1} = 0$ and $\left\|h_\eta^{s-1}\right\|_2 / \sqrt{N} \le 1 + \tau_0$,

$$e_s \le 2e^{-A/2} + (8/A)\rho\sqrt{C}(1+\tau_0) + (8b_L/A)e_{s-1} + \tau_0 \le e_*.$$

Each term is at most $e_*/4$. In the first layer, the preactivation perturbation is at most $\rho(d+1) \le A/4$, which initializes the same invariant. At the readout,

$$f(x)z_w(x;\eta) \ge 15a/16 - \rho(1+\tau_0) - \tau_0 > 1.$$

Furthermore, $\sqrt{N}\,\|v_*\|_2 \ge a\sqrt{C/2}$ and $\sqrt{N}\,\|v - v_*\|_2 \le \rho$, so

$$E_v \ge a^2 C/8 \ge 2C, \qquad r_c(w) \le \frac{2\sqrt{2}c}{a\sqrt{C}} \le \tau_0.$$

Thus all noisy and clean clipped outputs on the calibrated ball equal $f(x)/2$, establishing membership in $\mathcal{R}_{N,c}$.

The same clone projections in (1) map this cylinder back into itself. Averaging rows preserves the coordinate intervals because the teacher rows agree within each group. Double projection kills the bulk, since $E^s P = 0$. The first-layer and readout intervals are likewise preserved. The averaged network therefore also has signed margin greater than 1 on every input. Its clipped predictions agree exactly with the original network, and its population count is C. Moreover,

$$\sqrt{E_v} \ge a\sqrt{C/2} - \rho \ge (5/8)a\sqrt{C}, \qquad M(w) \le \|v\|_1 \le a + \rho \le 5a/4.$$

Thus $(C-1)M(w)^2 \le 4E_v$, proving membership in $\mathcal{A}_N$.

There are Nq_0 independent constrained scalar coordinates; the sum of squares of their centers is at most NB_0. For $Z \sim \mathcal{N}(0, t^2)$, $\rho \le 1/4$, and any center m,

$$\mathbb{P}\{|Z - m| \le \rho t\} \ge \frac{2\rho}{e\sqrt{2\pi}} e^{-m^2/t^2}.$$

This follows by bounding the Gaussian density throughout the interval, using $(|m|/t + \rho)^2/2 \le m^2/t^2 + \rho^2$. Multiply the interval estimates and the bulk probability $1/2$. Since $N/t^2 \ge 1$ and $t \le 1$, the result is at least $\exp(-ND_*/t^2)$ with (6). □